1-Month Cybersecurity Jumpstart Plan

A structured daily roadmap to bridge the gap between academic theory and applied security.

Phase 1: Structure & Resources (Days 1-30)

For the next month, dedicate 1 to 1.5 hours daily. Consistency is more important than marathon study sessions. This month focuses on building foundational networking and command-line muscle memory before moving into applied scripting.

Recommended Free Platforms:

- **TryHackMe:** Excellent for guided, browser-based labs. Start with the *Pre-Security* learning path.
- **OverTheWire (Bandit):** The absolute best way to learn the Linux command line through hands-on wargames.
- **Professor Messer:** Free YouTube playlists for CompTIA Network+ and Security+ (essential for theoretical foundations).

Weekly Syllabus Breakdown

- **Week 1 (Networking & Packet Analysis):** The internet runs on TCP/IP. Focus on OSI layers, common protocols, VPN encapsulation (IPsec/IKE/ESP), and capturing raw traffic using Wireshark to perform basic forensics.
- **Week 2 (Linux CLI Mastery):** You cannot do cybersecurity efficiently on a graphical interface. Use OverTheWire to learn rapid navigation, file permissions, piping, and basic bash scripting.
- **Week 3 (Security Foundations & Threats):** Understand core vulnerabilities. Explore cryptography fundamentals and introductory web vulnerabilities, keeping in mind how defensive solutions (like firewalls) would intercept them.
- **Week 4 (Applied Python Projects):** Bridge programming and security. Write custom Python tools—such as an automated port scanner or a steganography script to hide data in images. Document this process extensively to build out your portfolio.

Phase 2: GitHub Repository Setup

Your GitHub is your public proof of competence. Keep it clean and structured. Create a repository named `cyber-100-days` or `security-portfolio`.

Recommended Folder Structure

```
security-portfolio/
│
├── week-01-networking/
│   ├── pcap-analysis-lab.md
│   └── vpn-architecture-notes.md
│
├── week-02-linux/
│   └── bandit-wargame-writeup.md
│
├── projects/
│   └── python-steganography-tool/
│       ├── stego_script.py
│       └── README.md
│
└── README.md (Master Repo Description)
```

Phase 3: The Perfect README Template

Every time you finish a mini-project (like your Python scripts or a complex Wireshark analysis), include a README. Ensure you highlight the practical application of your work. Use this template:

```
# Project Name (e.g., Image Steganography Script)

## Objective
A brief 1-2 sentence description of what this project does and the problem it
solves.
*Example: A Python-based tool utilizing the Pillow library to encode and decode
hidden messages within PNG image files, demonstrating practical data obfuscation
techniques.*

## Tools & Technologies Used
- Python 3.x
- Libraries: PIL (Pillow), Cryptography (optional)
- OS: Ubuntu Linux (or Windows via WSL)

## Methodology / How It Works
1. **Encoding:** The script reads an image and converts a string of text into
binary.
2. **LSB Modification:** It modifies the Least Significant Bit of the image
pixels to embed the binary data without noticeably altering the image visually.
3. **Decoding:** It extracts the LSBs and converts the binary back into readable
text.

## Key Learnings
- Understanding how pixel data is stored and manipulated in arrays.
- Overcoming challenges with data loss during image compression.

## Usage
Provide the exact command line arguments to run your tool:
`python stego_script.py -e "Secret Message" -i input.png -o output.png`
```